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4.10 Using Resources

AQA GCSE Chemistry (8462) — Paper 2

Industries use the Earth's natural resources to manufacture useful products. In order to operate sustainably, chemists seek to minimise the use of limited resources, the use of energy, waste and environmental impact in the manufacture of these products. Chemists also aim to develop ways of disposing of products at the end of their useful life in ways that ensure that materials and stored energy are utilised. Pollution, disposal of waste products and changing land use have a significant effect on the environment, and environmental chemists study how human activity has affected the Earth's natural cycles. Sections marked **(Chemistry only)** are not in Combined Science; content marked **(HT only)** is Higher Tier.

4.10.1 Using the Earth's Resources and Obtaining Potable Water

4.10.1.1 Using the Earth's Resources and Sustainable Development

Humans use the Earth's resources to provide **warmth, shelter, food and transport**.

- **Natural resources**, supplemented by agriculture, provide **food, timber, clothing and fuels** (e.g. cotton, wool, wood, rubber).
- **Finite resources** from the Earth, oceans and atmosphere are processed to provide energy and materials (e.g. crude oil, metal ores, limestone).
- Chemistry plays an important role in improving agricultural and industrial processes to provide new products and in **sustainable development**.
- In some cases, **synthetic products** can **replace** or supplement natural products (e.g. synthetic rubber for natural rubber, polyester and nylon for wool and cotton, synthetic fertilisers for manure).

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Sustainable development is development that **meets the needs of current generations without compromising the ability of future generations to meet their own needs**.

Finite (non-renewable) resources will run out because they are used much faster than they are formed — e.g. fossil fuels, metal ores.
Renewable resources can be replaced at the same rate as they are used — e.g. timber, crops, fresh water.

Students should be able to state examples of natural products that are supplemented or replaced by agricultural and synthetic products, and distinguish between finite and renewable resources given appropriate information. They should also be able to extract and interpret information about resources from charts, graphs and tables, and use orders of magnitude to evaluate the significance of data.

Exam Tip: The AQA definition of sustainable development is examined word-for-word. Learn it exactly: "meeting the needs of current generations without compromising the potential of future generations to meet their own needs."

4.10.1.2 Potable Water

Potable water is water that is **safe to drink**. It is **not pure water** in the chemical sense because it contains **dissolved substances**.

Water of appropriate quality is essential for life. For humans, drinking water should have **sufficiently low levels of dissolved salts and microbes**.

Exam Tip: "Potable water is pure water" is **wrong**. Potable water contains dissolved salts (e.g. calcium and magnesium ions, small amounts of chlorine) — it is a **mixture**. Pure water in chemistry means only H_2O molecules.

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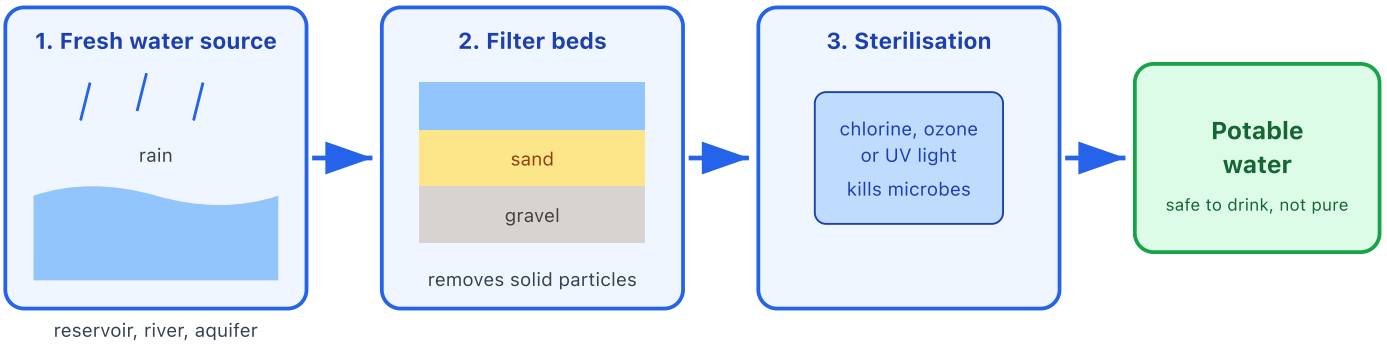
Methods to Produce Potable Water

Methods used depend on **available supplies** of water and **local conditions**.

In the United Kingdom (UK), **rain** provides water with **low levels of dissolved substances** (fresh water) that collects in the ground and in lakes and rivers. Most potable water is produced by:

1. **Choosing an appropriate source of fresh water** — ground water from aquifers (porous rock), reservoirs, lakes or rivers.
2. **Passing the water through filter beds** — to remove any solids (leaves, twigs, sand, grit).
3. **Sterilising** — to kill microbes. Sterilising agents used for potable water include **chlorine**, **ozone** or **ultraviolet light**.

Producing potable water from fresh water (UK)



Desalination of Sea Water

If supplies of fresh water are limited (e.g. in dry countries such as those in the Middle East), **desalination** of salty water or sea water may be required. Desalination can be done by:

Method	How it works	Disadvantages
Distillation	Sea water is boiled; the water evaporates leaving the salts behind; the steam is condensed to give pure water.	Requires **large amounts of energy** to boil the water — expensive; usually uses fossil fuels.
Reverse osmosis	Sea water is forced under high pressure through a **membrane** that lets water	Requires **large amounts of energy** to create the high pressure;

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	molecules pass through but holds back the dissolved salts.	membranes are expensive and must be replaced.

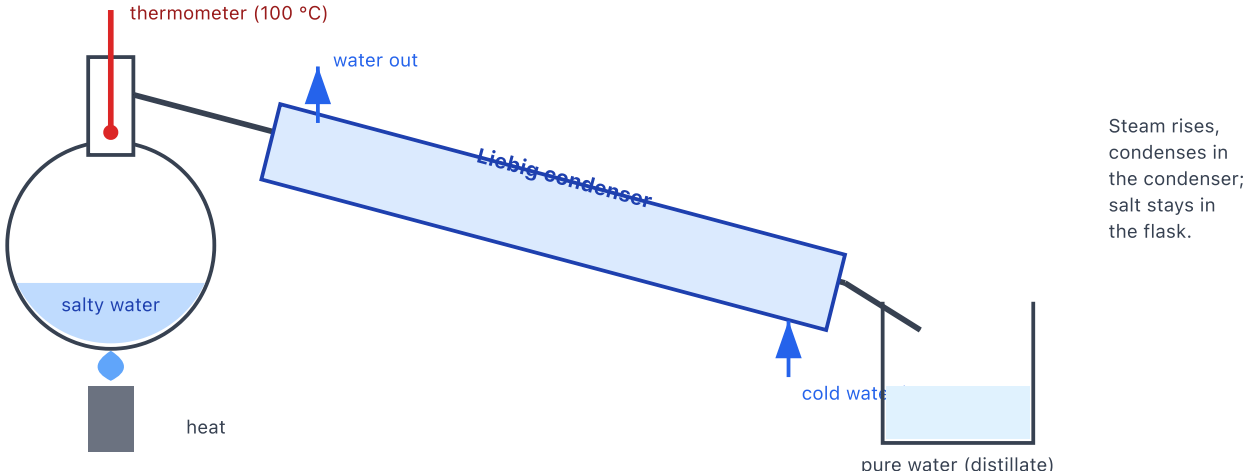
Both desalination processes are therefore expensive and are only used where fresh water is scarce. The water produced may need minerals added back and sterilisation before drinking.

Required Practical — Analysis and Purification of Water Samples

Aim: analyse and purify water samples from different sources, including **pH**, **dissolved solids** and **distillation**.

1. **pH:** test each sample with universal indicator (compare with colour chart) or a calibrated **pH meter**.
2. **Mass of dissolved solids:** weigh an empty evaporating basin; add a measured volume (e.g. 10 cm^3) of the water; heat gently to evaporate the water; reweigh. The increase in mass is the mass of dissolved solids.
3. **Distillation:** heat salty water in a flask; the steam passes into a **condenser** (cold water flowing in at the bottom and out at the top) and the condensed pure water is collected. Test the distillate: its pH should be 7, it should leave no residue when evaporated, and it should boil at 100 °C.

Simple distillation of salty water



Exam Tip: In the distillation set-up, cold water enters the condenser at the **bottom** (lower end) and leaves at the **top**, so the condenser jacket stays completely full of cold water. The

> thermometer bulb is at the side-arm, measuring the temperature of the vapour — 100 °C confirms the distillate is water.

Worked Example 1 — Dissolved Solids

```
Question: 25.0 cm³ of a water sample was evaporated to dryness.
          Mass of empty basin = 41.62 g
          Mass of basin + residue = 41.67 g
          Calculate the concentration of dissolved solids in g/dm³.

Mass of solids = 41.67 − 41.62 = 0.05 g
Volume         = 25.0 cm³ = 25.0 ÷ 1000 = 0.0250 dm³
Concentration  = mass ÷ volume = 0.05 ÷ 0.0250 = 2.0 g/dm³

Answer:  2.0 g/dm³
(Note: such a small mass difference has a large percentage
 uncertainty — using a larger volume would improve accuracy.)
```

4.10.1.3 Waste Water Treatment

Urban lifestyles and industrial processes produce large amounts of **waste water** that require treatment before being released into the environment.

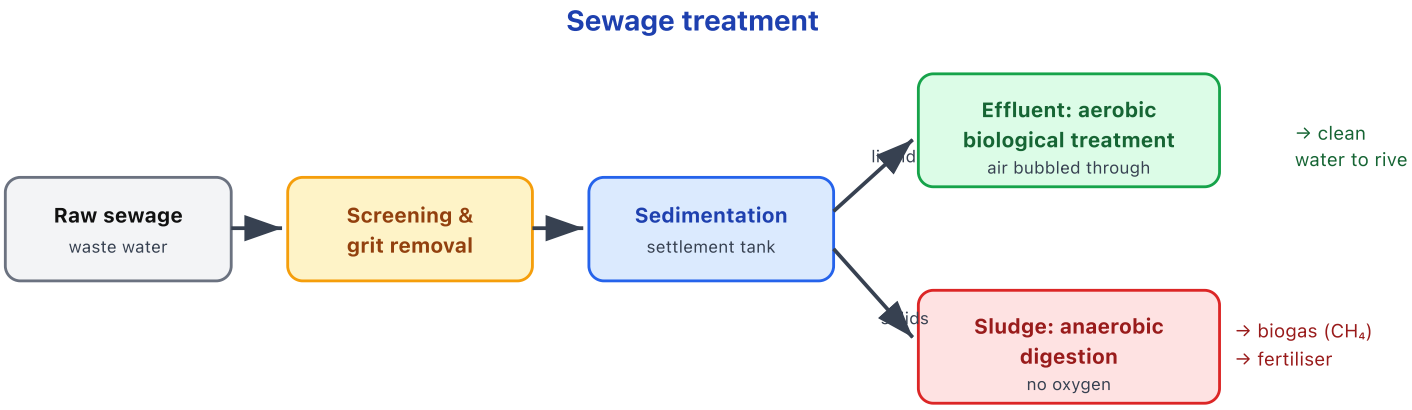
- **Sewage** and **agricultural waste water** require removal of **organic matter** and **harmful microbes**.
- **Industrial waste water** may require removal of **organic matter** and **harmful chemicals** (e.g. heavy metals, detergents).

Sewage treatment includes:

1. **Screening and grit removal** — large objects (rags, plastics, twigs) and grit/sand are removed.
2. **Sedimentation** — in large settlement tanks, heavier solids sink to produce **sewage sludge**; the lighter liquid on top is called **effluent**.
3. **Anaerobic digestion of sewage sludge** — bacteria break down the organic matter in the sludge **without oxygen**. This produces **biogas** (mainly methane) that can be burned to generate electricity, and digested waste that can be used as **fertiliser**.

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4. **Aerobic biological treatment of effluent** — air is bubbled through the effluent so that aerobic bacteria break down the remaining organic matter and microbes. The clean water can then be released into rivers or the sea.



Industrial waste water may also need extra stages to remove harmful chemicals.

Comparing Sources of Potable Water

Source	Processes needed	Energy / cost
Ground water / fresh water	Filtration and sterilisation	Lowest — the easiest to treat
Waste water	Screening, sedimentation, biological treatment, then further treatment and sterilisation	Medium — more stages; public may be reluctant to drink recycled sewage water
Salt water (sea water)	Desalination (distillation or reverse osmosis), then treatment	Highest — huge energy demand

Exam Tip: Students should be able to comment on the relative ease of obtaining potable water from **waste, ground and salt water**. The usual ranking is ground water easiest, then waste water, then salt water (most energy-intensive). Always justify with *energy* and *number of processes*.

4.10.1.4 Alternative Methods of Extracting Metals (HT only)

The Earth's resources of metal ores are **limited**. Copper ores are becoming scarce, and new ways of extracting copper from **low-grade ores** (ores containing only small amounts of copper) are being researched to limit the environmental impacts of traditional mining (digging, moving and disposing of large amounts of rock).

Copper can be extracted from low-grade ores using biological methods:

Method	How it works
Phytomining	Uses **plants** to absorb metal compounds from the soil. The plants are **harvested and then burned** to produce **ash** that contains metal compounds. The metal compounds are then processed to obtain the metal.
Bioleaching	Uses **bacteria** to produce **leachate solutions** that contain metal compounds (e.g. copper ions in solution).

The metal compounds obtained by phytomining and bioleaching can then be processed to obtain the metal. For example, copper can be obtained from solutions of copper compounds by:

- **Displacement using scrap iron** (iron is more reactive than copper): $Fe + CuSO_4 \rightarrow FeSO_4 + Cu$
  Ionic: $Fe(s) + Cu^{2+}(aq) \rightarrow Fe^{2+}(aq) + Cu(s)$
- **Electrolysis** — copper is deposited at the negative electrode (cathode): $Cu^{2+} + 2e^- \rightarrow Cu$

	Advantages	Disadvantages
Phytomining	Extracts metal from low-grade ores and contaminated land; less damage than mining; less waste rock; plants are renewable; can clean up polluted soil	**Very slow** — plants take time to grow; depends on weather/seasons
Bioleaching	Uses low-grade ores; less energy than smelting; no need to dig/move large amounts of rock	**Slow**; leachate can be **toxic** and acidic if it escapes into the environment

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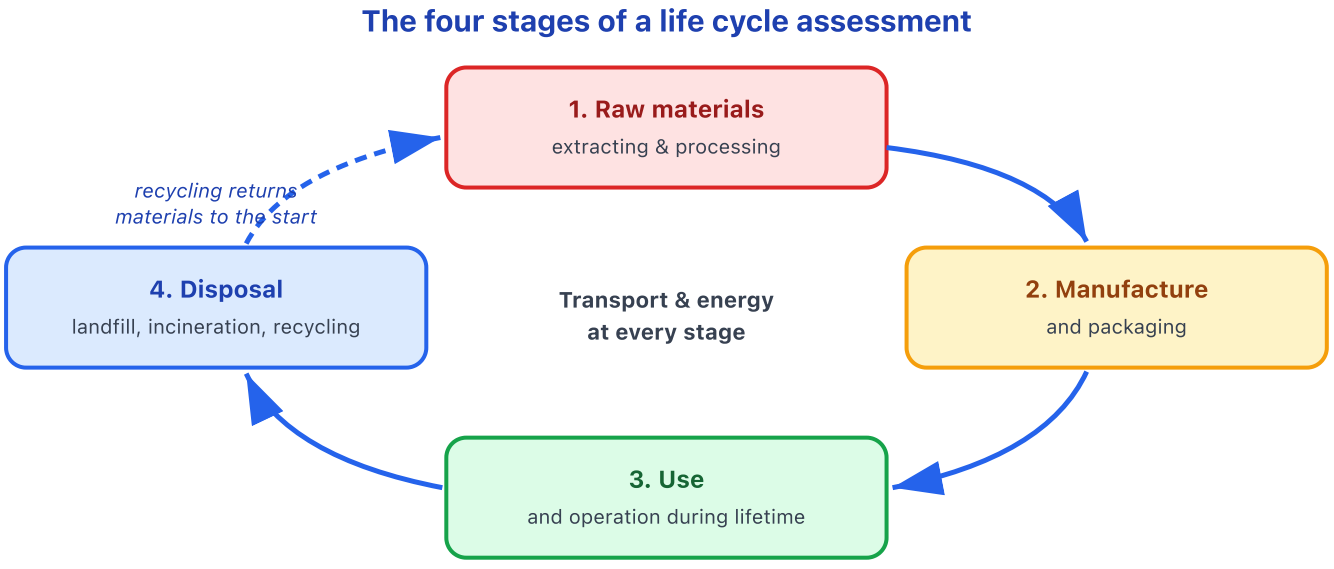
Exam Tip: The examiner wants you to say these methods **avoid traditional mining**, which involves "digging, moving and disposing of large amounts of rock". Both methods are slow, so their main justification is that high-grade copper ores are running out. Remember the final step: scrap iron **displacement** or **electrolysis**.

4.10.2 Life Cycle Assessment and Recycling

4.10.2.1 Life Cycle Assessment

A **life cycle assessment (LCA)** is carried out to assess the **environmental impact** of products in each of these stages:

- **extracting and processing raw materials**
- **manufacturing and packaging**
- **use and operation during its lifetime**
- **disposal at the end of its useful life**, including transport and distribution at each stage.



At each stage, consider: use of water, resources, energy sources, and production of waste/pollution.

Limitations of LCAs

- **Use of water, resources, energy sources and production of some wastes** can be fairly easily **quantified** (measured with numbers).
- **Allocating numerical values to pollutant effects is less straightforward** and requires **value judgements**, so LCA is not a purely objective process.
- **Selective or abbreviated LCAs** can be devised to evaluate a product, but these can be **misused** to reach pre-determined conclusions, e.g. in support of claims for **advertising purposes**.

Comparing Shopping Bags — an Example LCA

Stage	Plastic (polythene) carrier bag	Paper carrier bag
Raw materials	Crude oil (finite) — fractional distillation, cracking, polymerisation	Timber (renewable) — but trees must be felled and pulped
Manufacture	Lower energy; little waste; bags are light	Pulping and processing use lots of energy and water; produce lots of waste
Use	Strong; can be reused many times; waterproof	Usually used only once; weak when wet
Disposal	Not biodegradable — persists in landfill for decades; litter harms wildlife; can be recycled	Biodegradable; non-toxic; easily recycled

Worked Example 2 — Interpreting LCA Data

```
Data per bag:                          Plastic bag    Paper bag
Raw materials                          crude oil      timber
Fossil-fuel energy to make (MJ)            1.6            1.7
Solid waste (g)                             14             50
CO₂-equivalent emissions (g)               230            530
Fresh water used (litres)                  0.1            3.3

Question: Use the data to compare the two bags.
```

```
Answer:  The paper bag produces over 3 times more solid waste
         (50 g vs 14 g), more than twice the greenhouse-gas
         emissions (530 g vs 230 g) and uses 33 times more water.
         Energy use is similar. However, the paper bag is made from
         a renewable resource and is biodegradable, whereas the
         plastic bag uses finite crude oil and persists for many
         years. A reusable plastic bag used many times is likely
         to have the lowest impact per use.

(Figures are illustrative of typical published comparisons.)
```

Exam Tip: LCA questions nearly always give you a data table. Make **direct numerical comparisons** ("x is three times higher than y"), cover **all four stages**, and finish with a **conclusion**. For "evaluate" questions, also mention a **limitation** — e.g. the values for pollution involve value judgements, or the LCA may have been commissioned by a company with a vested interest.

4.10.2.2 Ways of Reducing the Use of Resources

The reduction in use, reuse and recycling of materials by end users **reduces the use of limited resources**, **use of energy sources**, **waste** and **environmental impacts**.

- **Metals, glass, building materials, clay ceramics and most plastics** are produced from **limited raw materials**.
- Much of the **energy** for the processes comes from **limited resources** (fossil fuels).
- **Obtaining raw materials from the Earth by quarrying and mining causes environmental impacts** — noise, dust, destruction of habitats, visual scarring, waste heaps.

Material	Reuse / recycling method
Glass	Glass bottles can be **reused** (washed and refilled). Glass that is not reused is **crushed and melted** to make different glass products. Products that do not need clear glass (e.g. green bottles) can be made from mixed glass.
Metals	Recycled by **melting and recasting or reforming** into different products. The amount of separation required depends on the metal and the product — e.g. scrap steel can

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	be added to iron from a blast furnace to reduce the amount of iron that needs to be extracted from iron ore.
Plastics	Sorted by type, washed, shredded, melted and remoulded (thermosoftening plastics only).

Advantages and Disadvantages of Recycling

Advantages	Disadvantages
Conserves finite raw materials (ores, crude oil, sand)	Collection and transport of waste uses energy and costs money
Uses **less energy** than extraction (recycling aluminium uses about 5% of the energy needed to extract it from bauxite)	Sorting materials is labour-intensive and can be expensive
Less mining/quarrying — less environmental damage	Quality of recycled material may be lower (e.g. some plastics degrade)
Less waste going to landfill	Some materials (e.g. mixed composites) are hard to separate

4.10.3 Using Materials (Chemistry only)

4.10.3.1 Corrosion and Its Prevention

Corrosion is the **destruction of materials by chemical reactions with substances in the environment**.

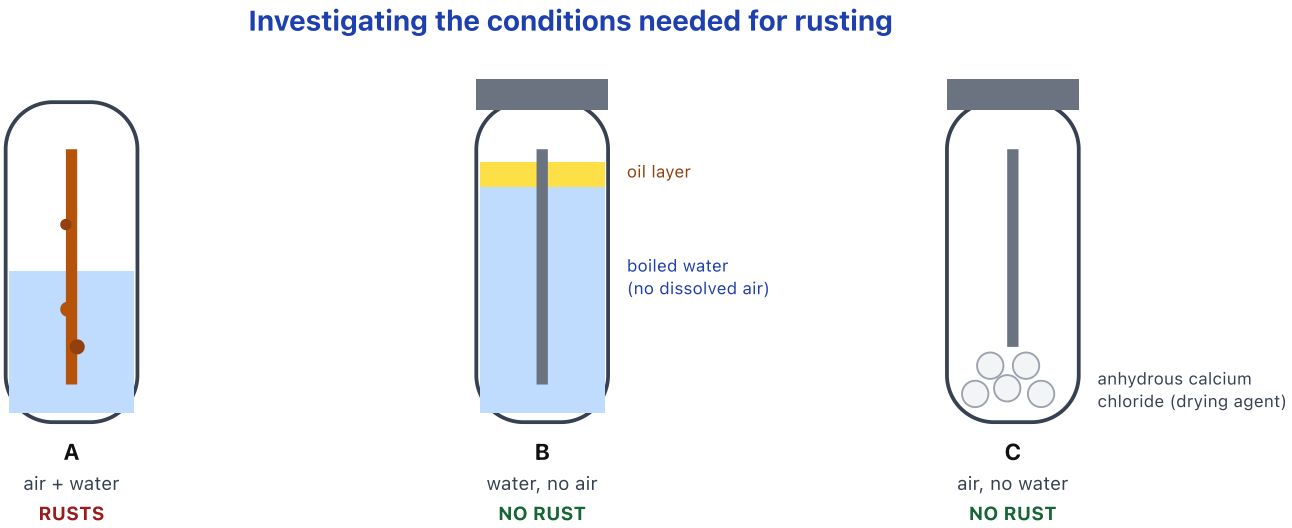
Rusting is an example of corrosion. **Both air (oxygen) and water are necessary for iron to rust.**

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iron + oxygen + water → hydrated iron(III) oxide (rust)
$4Fe(s) + 3O_2(g) + 2xH_2O(l) \rightarrow 2Fe_2O_3 \cdot xH_2O(s)$

Rusting is an oxidation reaction — iron atoms lose electrons:
$Fe \rightarrow Fe^{3+} + 3e^-$

Experiment to Show That Air and Water Are Needed



Boiling the water in tube B removes dissolved air; the oil layer stops air re-dissolving. Calcium chloride in tube C absorbs water vapour from the air. Only nail A, with **both** air and water, rusts.

Preventing Corrosion

Corrosion can be prevented by applying a **coating that acts as a barrier**, or by **sacrificial protection**.

Method	How it works	Examples
Barrier methods — **greasing, painting, electroplating**, oiling, plastic coating	Stop **air and/or water** reaching the metal surface. If the barrier is scratched, the exposed metal will corrode.	Painting car bodies and bridges; greasing bicycle chains; electroplating tin onto steel cans; chromium plating

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Sacrificial protection	A **more reactive metal** is attached to or coats the iron. The more reactive metal **loses electrons (oxidises) in preference** to iron, so it corrodes instead of the iron. Works even if the surface is scratched.	Blocks of **magnesium or zinc** bolted to ships' hulls, pipelines and oil rigs
Galvanising	Coating iron/steel with a layer of **zinc**. Acts as **both** a barrier **and** sacrificial protection — if scratched, the zinc still corrodes in preference to the iron.	Buckets, nails, fences, car bodies, lamp posts

Aluminium is a reactive metal but does not corrode significantly, because it has a thin **oxide coating** (Al_2O_3) that forms naturally and protects the metal from further corrosion.

> **Exam Tip:** To explain sacrificial protection you must use reactivity and electrons: "Zinc is **more reactive** than iron, so zinc **loses electrons** / is **oxidised** in preference to iron." Saying "zinc rusts instead" is wrong — only iron *rusts*; other metals *corrode*.

4.10.3.2 Alloys as Useful Materials

Most metals in everyday use are **alloys** — mixtures of a metal with other elements (usually other metals or carbon). Pure metals are often too soft for many uses because their layers of atoms slide over each other easily. In an alloy, the different-sized atoms **distort the layers**, making it **harder** for them to slide, so alloys are **harder and stronger**.

Alloy	Composition	Properties	Uses
Bronze	Copper and **tin**	Hard, strong, resistant to corrosion	Statues, medals, bells, ship propellers
Brass	Copper and **zinc**	Harder than copper, workable, gold-coloured	Water taps, door fittings, musical instruments, electrical plugs

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Gold alloys	Gold with silver, copper and zinc	Harder than pure gold	Jewellery
High carbon steel	Iron with a higher proportion of carbon	**Strong but brittle**	Cutting tools, blades, drill bits
Low carbon steel (mild steel)	Iron with a low proportion of carbon	**Softer and more easily shaped**	Car bodies, girders, wire, food cans
Stainless steel	Iron with **chromium and nickel** (and carbon)	**Hard and resistant to corrosion**	Cutlery, sinks, surgical instruments, cookware
Aluminium alloys	Aluminium with e.g. copper, magnesium	**Low density**, strong	Aircraft, bicycles, high-performance cars

Carats — Measuring the Purity of Gold

The proportion of gold in an alloy is measured in **carats**. **24 carat is 100% (pure) gold**.

`percentage of gold = (carat number ÷ 24) × 100`

So **18 carat gold is 75% gold** (18 ÷ 24 = 0.75).

Worked Example 3 — Carats

```
Question: (a) What percentage of a 9 carat gold ring is gold?
          (b) The ring has a mass of 4.8 g. What mass of gold
              does it contain?

(a) % gold = (9 ÷ 24) × 100 = 37.5%

(b) mass of gold = 4.8 × 0.375 = 1.8 g

Answer:  (a) 37.5%    (b) 1.8 g
```

> **Exam Tip:** You will not be asked to recall the composition of every alloy, but you should know **bronze = copper + tin**, **brass = copper + zinc**, **stainless steel contains chromium and nickel**, and the difference between high and low carbon steels. Questions often give you data on unfamiliar alloys and ask you to choose one for a given use — match a *property* to the *use* every time.

4.10.3.3 Ceramics, Polymers and Composites

Glass

Most of the glass we use is **soda-lime glass**, made by heating a mixture of **sand, sodium carbonate and limestone**. **Borosilicate glass**, made from **sand and boron trioxide**, **melts at higher temperatures** than soda-lime glass, so it is used for laboratory glassware and ovenware (e.g. Pyrex).

Clay Ceramics

Clay ceramics, including **pottery and bricks**, are made by **shaping wet clay and then heating it in a furnace** (a kiln). They are hard, brittle, resistant to heat and electrical insulators.

Polymers

The **properties of polymers depend on what monomers they are made from and the conditions under which they are made**.

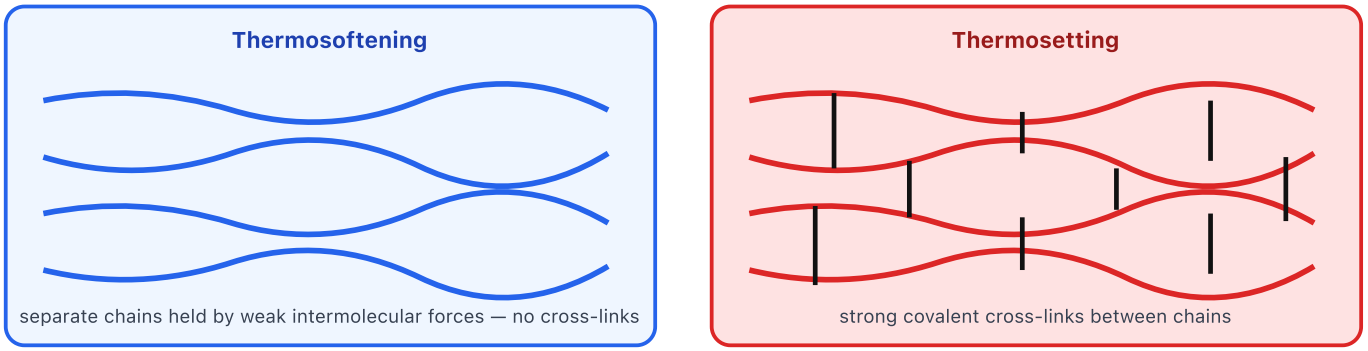
	Low density poly(ethene), LDPE	High density poly(ethene), HDPE
Monomer	Ethene	Ethene
Conditions	Very **high pressure** (~1000–3000 atm), ~200 °C, trace of oxygen	**Lower temperature and pressure** with a **catalyst** (Ziegler–Natta)
Structure	Highly **branched** chains — cannot pack closely	Mostly **straight** chains — pack closely together

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Properties	Flexible, lower density, lower softening temperature	Rigid, stronger, higher density, higher softening temperature
Uses	Carrier bags, cling film, squeezy bottles	Water pipes, milk bottles, crates

Thermosoftening and Thermosetting Polymers

Thermosoftening vs thermosetting polymers



	Thermosoftening polymers	Thermosetting polymers
Structure	**Individual, tangled polymer chains** with only weak intermolecular forces between them	Chains joined by strong covalent **cross-links**
Effect of heat	**Melt / soften when heated** (intermolecular forces overcome) — can be remoulded and recycled	**Do not melt when heated** — they char/decompose at high temperature
Examples	Poly(ethene), poly(propene), PVC	Melamine, Bakelite, epoxy resins, vulcanised rubber
Uses	Bottles, bags, pipes, packaging	Saucepan handles, electrical plug sockets, kitchen worktops

Composites

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Most **composites** are made of two materials: a **matrix or binder** surrounding and binding together **fibres or fragments** of the other material, which is called the **reinforcement**. The properties of a composite combine the useful properties of its components.

Composite	Reinforcement	Matrix / binder	Properties and uses
Fibreglass (glass-reinforced plastic)	Glass fibres	Polymer resin	Low density, strong, can be moulded — boats, skis, surfboards
Carbon fibre composite	Carbon fibres (or carbon nanotubes)	Polymer resin	Very strong and light — racing cars, bicycles, aircraft parts, tennis rackets
Concrete	Sand and gravel (aggregate)	Cement	Strong in compression — buildings, bridges (reinforced concrete adds steel rods for strength in tension)
Wood (natural composite)	Cellulose fibres	Lignin	Strong, renewable — construction, furniture

Comparing Material Types

Material	Typical properties
Metals	Good conductors of heat and electricity; malleable; strong; often dense; may corrode
Ceramics (glass, clay)	Hard, brittle, insulators, high melting points, resistant to corrosion
Polymers	Low density, insulators, flexible (thermosoftening) or rigid (thermosetting), easily moulded; usually not biodegradable
Composites	Tailored combination of properties — e.g. high strength with low density

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> **Exam Tip:** Students should be able to compare quantitatively the physical properties of glass and clay ceramics, polymers, composites and metals, and explain how the properties of materials are related to their uses — and to select appropriate materials. In a "suggest which material" question, quote the relevant data from the table you are given (e.g. density, strength, cost) and link each property to a requirement of the use.

4.10.4 The Haber Process and the Use of NPK Fertilisers (Chemistry only)

4.10.4.1 The Haber Process

The **Haber process** is used to manufacture **ammonia**, which can be used to produce **nitrogen-based fertilisers**.

> `nitrogen + hydrogen ⇌ ammonia`
>
> $N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$
>
> The reaction is **reversible**. The forward reaction is **exothermic** ($\Delta H = -92$ kJ/mol).

Raw Materials

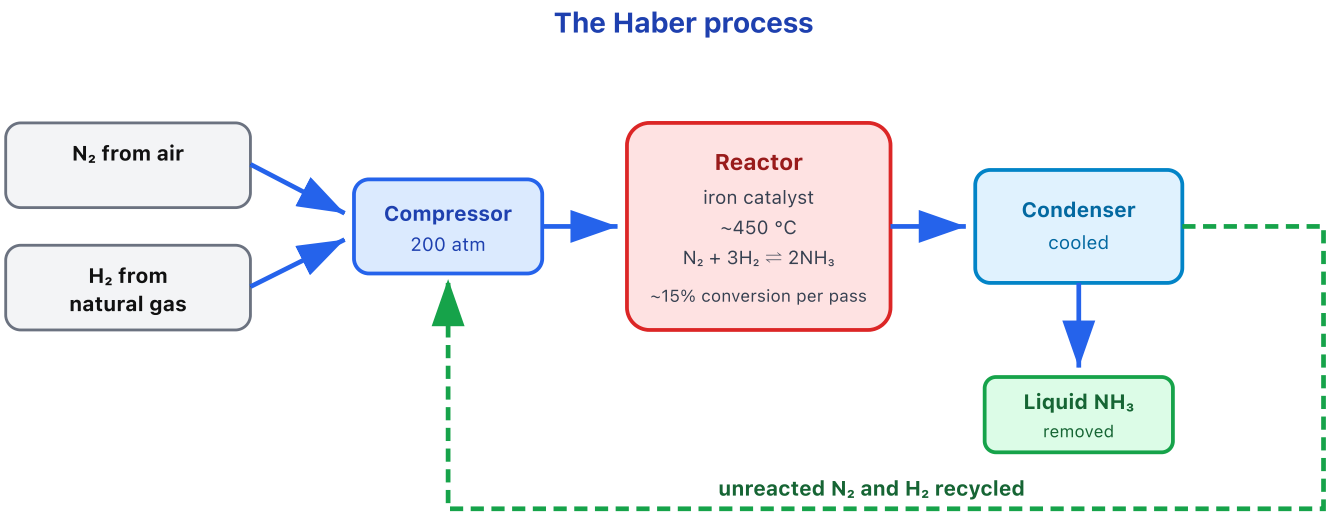
- **Nitrogen** is obtained from the **air** (fractional distillation of liquefied air).
- **Hydrogen** may be obtained from **natural gas** (methane) or other sources, e.g. by reacting methane with steam: $CH_4 + H_2O \rightarrow CO + 3H_2$.

The Process

1. The purified gases are passed over a **catalyst of iron** at a **high temperature (about 450 °C)** and a **high pressure (about 200 atmospheres)**.
2. Some of the hydrogen and nitrogen reacts to form ammonia. The reaction is reversible, so some of the ammonia produced breaks down into nitrogen and hydrogen.

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3. On cooling, the **ammonia liquefies and is removed** (ammonia has a much higher boiling point than N_2 and H_2).
4. The remaining **unreacted hydrogen and nitrogen are recycled** back into the reactor.



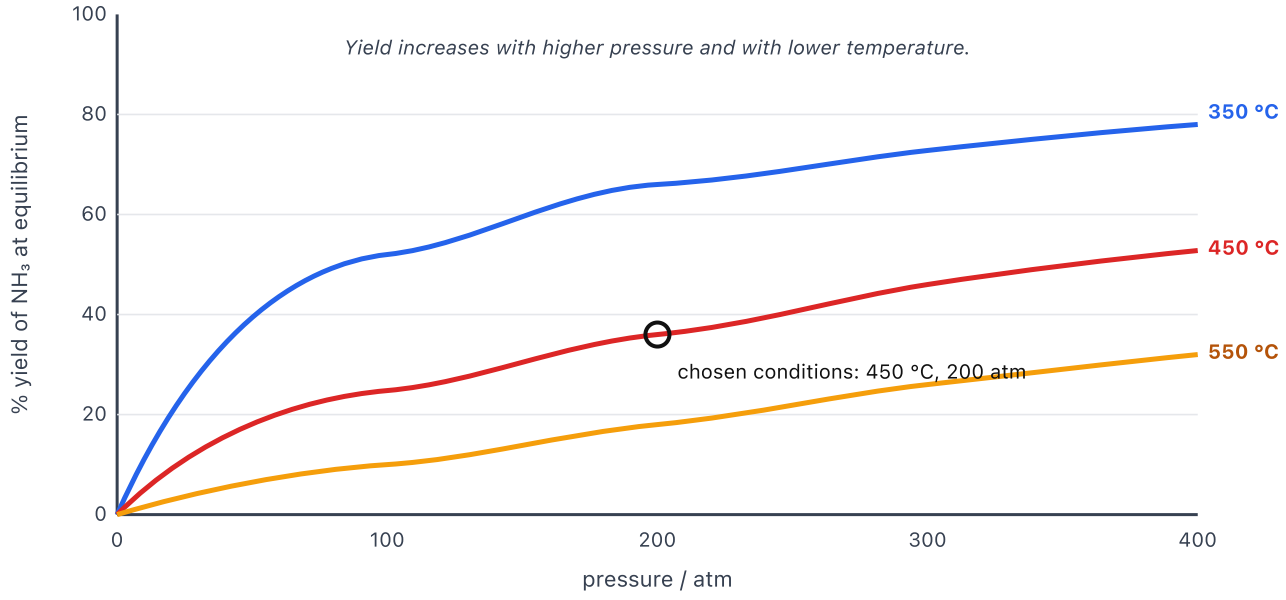
Choosing the Conditions (HT only)

Students should be able to interpret graphs of reaction conditions versus rate and apply the principles of dynamic equilibrium (from 4.6.2) to explain how the commercially used conditions are a **compromise** between **rate**, **equilibrium position (yield)** and **cost/energy**.

Condition	Effect on equilibrium yield	Effect on rate	Other considerations	Compromise chosen
Temperature	Forward reaction is **exothermic**, so a **lower** temperature shifts the equilibrium to the right → **higher yield**	Lower temperature → **slower** rate (fewer successful collisions; catalyst less effective)	Too slow at low temperature to be economic	**~450 °C** — reasonable yield at an acceptable rate
Pressure	4 molecules of gas on the left, 2 on the right, so	Higher pressure → gas particles closer	Very high pressures are **expensive** (strong	**~200 atm** — high yield and rate without

	higher pressure shifts equilibrium to the side with **fewer molecules** (right) → **higher yield**	together → more frequent collisions → **faster**	pipes and vessels, energy for compressors) and **dangerous**	excessive cost or risk
Catalyst (iron)	**No effect** on the position of equilibrium or the yield	**Increases** the rate of both forward and reverse reactions equally — equilibrium is reached faster	Allows a lower temperature to be used economically	Iron catalyst used

Equilibrium yield of ammonia vs pressure (approximate)



Exam Tip: For a 6-mark Haber question, structure your answer as **temperature → pressure → catalyst**, and for each one state the effect on **yield** (with the equilibrium reason) *and* on **rate**, then explain the **compromise**. Key phrases: "exothermic forward reaction", "fewer moles/molecules of gas on the right", "a catalyst does not change the yield". Mention that

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recycling unreacted gases gives an overall conversion of about 97% even though each pass converts only about 15%.

4.10.4.2 Production and Uses of NPK Fertilisers

Compounds of **nitrogen, phosphorus and potassium** are used as fertilisers to **improve agricultural productivity**. Plants need these elements for healthy growth, and crops remove them from the soil each year.

NPK fertilisers contain compounds of all three elements — **N**itrogen, **P**hosphorus and **K** (potassium) — in appropriate **percentages**. They are **formulations** of various salts containing appropriate percentages of the elements.

Element	Needed for	Source used in fertilisers
Nitrogen (N)	Making proteins; healthy leaf growth	**Ammonia** from the Haber process → ammonium salts and nitrates (e.g. ammonium nitrate, ammonium sulfate)
Phosphorus (P)	Root growth; DNA and energy transfer	**Phosphate rock**, which is mined and then treated with acids
Potassium (K)	Fruit and flower growth; resistance to disease	**Potassium chloride** and **potassium sulfate**, which are **mined** and can be used directly

Industrial Production of Fertiliser Salts

Ammonia can be used to manufacture **ammonium salts** and **nitric acid**:

```
Nitric acid is made by oxidising ammonia (several stages):
   ammonia + oxygen → ... → nitric acid

Ammonium nitrate (nitrogen fertiliser):
   ammonia + nitric acid → ammonium nitrate
   NH₃ + HNO₃ → NH₄NO₃

Ammonium sulfate:
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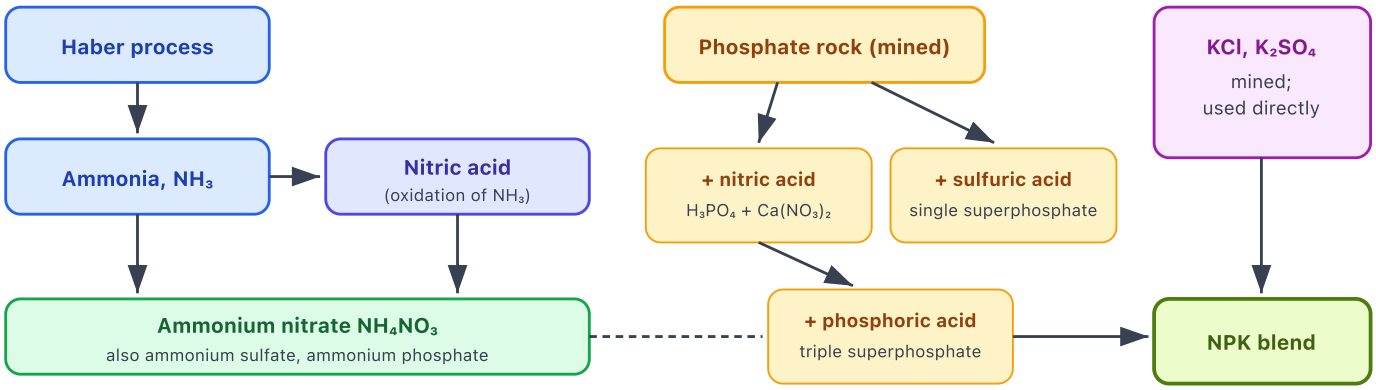
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    ammonia + sulfuric acid → ammonium sulfate
    2NH₃ + H₂SO₄ → (NH₄)₂SO₄

Ammonium phosphate:
    ammonia + phosphoric acid → ammonium phosphate
    3NH₃ + H₃PO₄ → (NH₄)₃PO₄
```

Phosphate rock is insoluble, so it cannot be used directly as a fertiliser — plants can only absorb soluble compounds. It is treated with acids:

Phosphate rock treated with...	Products
Nitric acid	**Phosphoric acid** and **calcium nitrate**
Sulfuric acid	**Single superphosphate** — a mixture of **calcium phosphate and calcium sulfate**
Phosphoric acid	**Triple superphosphate** (calcium phosphate)

Making the components of NPK fertilisers



Laboratory vs Industrial Production of Ammonium Sulfate

Students should be able to compare the industrial production of fertilisers with laboratory preparations of the same compounds.

Feature	Laboratory preparation	Industrial production

Raw materials	Dilute **ammonia solution** and dilute **sulfuric acid** bought from suppliers	Raw materials made on site: ammonia from the Haber process; sulfuric acid from sulfur (Contact process)
Method	**Titration** to find exact volumes, then mix, **evaporate** some water and leave to **crystallise**	Reactants react in a large reaction vessel; the hot concentrated solution is sprayed into a tower to dry and crystallise
Scale	**Small scale** — a few grams	**Very large scale** — thousands of tonnes per day
Process type	**Batch**	**Continuous**
Number of stages	Few (one reaction)	**Several stages** (making ammonia, making sulfuric acid, then neutralisation)
Concentration of reactants	Dilute	Concentrated
Energy	Heating needed to evaporate water	Heat released by the exothermic reaction is used to evaporate water

Worked Example 4 — Percentage of Nitrogen in a Fertiliser

Question: Compare the percentage by mass of nitrogen in ammonium nitrate, NH_4NO_3, and ammonium sulfate, $(NH_4)_2SO_4$. (Ar: H = 1, N = 14, O = 16, S = 32)

NH_4NO_3: Mr = 14 + 4(1) + 14 + 3(16) = 80
mass of N = 2 × 14 = 28
% N = (28 ÷ 80) × 100 = 35%

$(NH_4)_2SO_4$: Mr = 2(14 + 4) + 32 + 4(16) = 132
mass of N = 2 × 14 = 28
% N = (28 ÷ 132) × 100 = 21.2%

Answer: Ammonium nitrate contains a much higher percentage of

```
nitrogen (35%) than ammonium sulfate (21.2%), so less
mass is needed to supply the same amount of nitrogen.
```

Exam Tip: Remember which raw material provides which element: **N** from the **Haber process** (ammonia), **P** from **phosphate rock** (treated with acid because the rock is insoluble), **K** from **mined** potassium chloride and potassium sulfate. The three phosphate-rock reactions are classic 1-mark recall questions — especially "single superphosphate = calcium phosphate + calcium sulfate".

Summary Checklist for 4.10 Using Resources

- State how humans use natural, agricultural, synthetic and finite resources; define sustainable development.
- Distinguish between finite and renewable resources and interpret data about resources.
- Define potable water and explain why it is not pure.
- Describe how potable water is produced in the UK (source, filtration, sterilisation with chlorine, ozone or UV).
- Describe desalination by distillation and reverse osmosis and explain why both need large amounts of energy.
- Describe the water analysis and purification required practical (pH, dissolved solids, distillation).
- Describe sewage treatment: screening, sedimentation, anaerobic digestion of sludge, aerobic treatment of effluent.
- Compare the ease of obtaining potable water from ground, waste and salt water.
- *HT:* describe phytomining and bioleaching and how copper is then obtained by displacement with scrap iron or electrolysis.
- Describe the four stages of a life cycle assessment; interpret LCA data; state the limitations of LCAs.
- Describe how glass and metals are reused and recycled, and evaluate recycling.
- *Chemistry only:* state the conditions for rusting; describe barrier methods, sacrificial protection and galvanising.

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- *Chemistry only:* describe the composition and uses of bronze, brass, gold alloys (carats), steels and aluminium alloys.
- *Chemistry only:* describe how soda-lime glass, borosilicate glass and clay ceramics are made; compare LDPE and HDPE; explain thermosoftening vs thermosetting polymers; describe composites.
- *Chemistry only:* describe the Haber process (raw materials, conditions, iron catalyst, recycling, removal of ammonia).
- *HT:* explain the choice of Haber process conditions as a compromise between yield, rate and cost.
- *Chemistry only:* describe the production of NPK fertilisers, including the reactions of phosphate rock with acids, and compare laboratory and industrial production of ammonium sulfate.

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